Ultrasonic Flow Meter Solutions

VORTEX

Time-Saving Installation.
Easy Flow Measurement.

Badger Meter

Ultrasonic Flow Meters

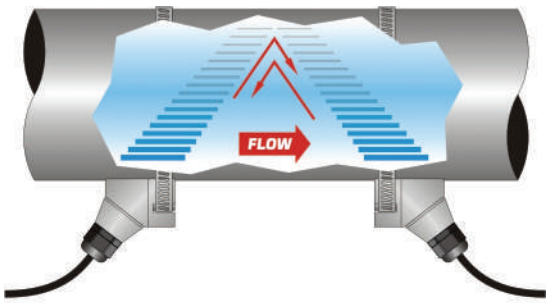
Badger Meter ultrasonic meters measure flow rate by propagating ultrasound waves into liquid-filled pipes and measuring the reflections. There are two technologies to choose from based on the fluid properties: transit time and Doppler. With clamp-on technology, the Badger Meter ultrasonic meters reside outside the pipe and bring you flow measurement with:

- Reduced installation costs
- Uninterrupted production
- Installation flexibility across a wide range of pipe sizes
- No pressure head loss
- No contact with internal liquid
- No moving parts to maintain

Transit Time

Clean fluids with small amounts of suspended solids or aeration

For clean fluids with small amounts of suspended solids or aeration, transit time ultrasonic meters provide the best performance and can be used in a wide range of applications for pipes 1/2 inch and larger.



Transit time flow meters have two transducers, which function as both ultrasonic transmitters and receivers. The flow meters operate by alternately transmitting and receiving a frequency modulated burst of ultrasound energy between the two transducers. The burst is first transmitted in the direction of fluid flow and then against fluid flow. Since ultrasound energy in a moving liquid is carried faster when it travels in the direction of fluid flow (downstream) than it does when it travels against fluid flow (upstream), a differential in the times of flight will occur. The ultrasound's time of flight is accurately measured in both directions, and the difference in time of flight is used to determine the velocity of the fluid.

- Ideally suited for permanent, temporary or portable flow verification
- Energy and network connectivity options available

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Transit Time Flow Meters

Transit time flow meters consist of electronics with remote or integral transducer options to measure bidirectional flow of clean liquids.

TFX-5000 Clamp-on Flow Meter
Supporting the widest range of pipe sizes and network and data logging options, TFX-5000 is our most versatile ultrasonic flow meter.
Pipe sizes: 1/2…48 inch (15…1200 mm)
Accuracy: up to ± 0.5% of reading

TFX-5000 Clamp-on Energy Meter
TFX-5000 meter with dual RTDs can calculate energy usage from the flow and temperature measurements across heating and cooling equipment and zones.
Pipe sizes: 1/2…48 inch (15…1200 mm)
Accuracy: up to ± 0.5% of reading

TFX-500w Clamp-on Flow Meter
TFX-500w flow meters are a cost-effective meter for measuring water flow in a variety of applications.
Pipe sizes: 1/2…10 inch (15…250 mm) or larger
Accuracy: ± 1% of reading

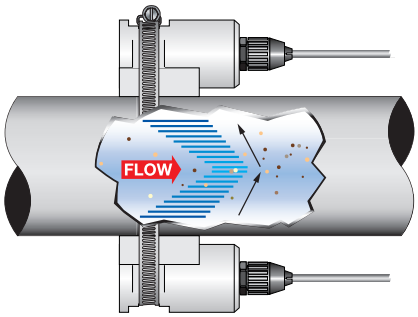
U500w In-Line Flow Meter
Simplified one-piece electronic meter and register integral to the meter body eliminates tampering, and, with no moving parts, also eliminates mechanical wear.
Pipe Sizes: 5/8...4 inch or larger
Accuracy: ± 1.5% of reading

Applications

- Water
- Water intake
- Pathogen control
- Corrosion control
- Pump stations
- Lift stations
- Boiler feed water
- Makeup water
- Heating/cooling water (glycol)
- Cooling towers
- Power generation
- Semiconductor
- Food and beverage

Doppler Flow Meters

Fluids with suspended solids
For fluids with significant amount of particles or air bubbles, such as slurries, sewage and plastics, Doppler ultrasonic meters provide the best performance.



Each Doppler flow meter utilizes two separate transducer heads. Ultrasonic waves are transmitted from one transducer and reflected by reflectors suspended within the liquid and then recorded by a receiving transducer. If the reflectors are moving within the ultrasound transmission path, ultrasound waves will be reflected at a frequency shifted (Doppler shift) from the transmitted frequency. The difference between the transmitted frequencies and reflected frequencies is directly proportional to the speed of the ultrasonic reflectors.

DFX Doppler Flow Meter
Designed for semi-clean liquids with suspended sonic reflectors on pipe sizes 1/4 inch (6 mm) and larger.

Accuracy: ± 2% full-scale over calibrated span

Clamp-on Transducers
Cable and flexible armored conduit selections are offered to reduce installation time.

- Ideal for most slurries or aerated liquids
- Low installation cost
- No moving parts to maintain

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Portable Flow Meters

Portable Flow Troubleshooting Kits
Designed for quick troubleshooting and verifying system performance during commissioning, portable ultrasonic kits are ideal tools throughout the process.

DXN Hybrid Flow and Energy Meter Kit
Our most versatile flow meter, the DXN measures flow with transit time and Doppler technology. The large, easy-to-read color display and intuitive touch-screen interface makes setup quick and easy. Measurements from external devices, such as RTDs and pressure transmitters, can be logged along with flow data. Accuracy: ± 1% of reading

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Open Channel Flow Meter

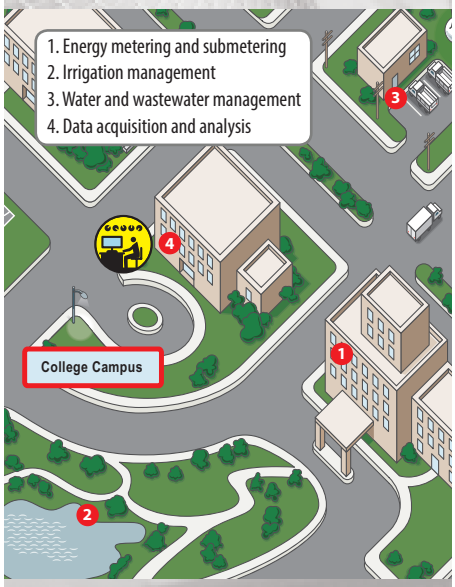
Measuring Open Flow
Badger Meter open channel flow meters use a non-contact ultrasonic level sensor to measure the water level in a flume, weir or other channel. Based on standard empirical equations, such as Gauckler–Manning, the flow rate can be calculated based on the dimensions and characteristics of the channel and the water level in the channel. This correspondence between water level and flow rate allows for a cost effective solution for measuring flow in an open channel or partially filled pipe.

iSonic 4000
Simple to install and setup, iSonic 4000 flow meters sends level, flow rate and total volume information over Modbus. The data logging feature provides a backup of measurements in case of network outages or reporting lapses.

Level sensor range: up to 49 in. (1.25 m)
Level accuracy: ± 0.125 in. (3 mm)

Applications

- Wastewater effluent
- Storm water
- Agriculture irrigation
- Industrial discharge



Communication Systems

Communication is Key
Connectivity is a key aspect to a successful flow monitoring system. Badger Meter ultrasonic meters adapt to almost any system utilizing communication networks, pulse, analog or wireless. We offer modular wireless components and data acquisition systems to provide greater flexibility, collect valuable operation data and allow remote monitoring.

Simple and Efficient
Our meters offer simple system performance solutions for flow measurement at an affordable cost. With a family of Badger Meter products for a wide range of needs, it is easy to go from a standalone system all the way to a full-service process environment.

- Flow/Btu monitors
- Btu energy transmitters
- Data acquisition systems
- AquaCUE® Flow Measurement Manager

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About Badger Meter

Badger Meter

Badger Meter Flow Instrumentation understands that companies cannot manage what they do not measure—and leverages more than a century of flow measurement expertise and a technology-rich portfolio to optimize customer applications worldwide.

An industry leader in both mechanical and electronic flow metering technologies, Badger Meter offers one of the broadest flow control and measurement portfolios in the industry—a portfolio that includes eight of the 10 major flow meter technologies. Simply put, Badger Meter Flow Instrumentation provides technology to measure and control whatever moves through a pipe or pipeline—including water, air, steam, oil, other liquids and gases.

Variety of Flow Instrumentation Solutions

M-Series® Mag Flow Meters

Industrial Oval Gear Flow Meters

Dynasonics® Ultrasonic Flow Meters

Research Control® Valves and Positioners

Hedland® Variable Area Flow Meters

Recordall® Disc Flow Meters

Impeller Flow Meters

Cox & Blancett® Turbine Flow Meters

Vortex Flow Meters

Preso® Differential Pressure Flow Meters

Flo-tech Hydraulic Fluid Testing

Coriolis Flow Meters

- Control
- Manage
- Optimize

Flow Dynamics® calibration services

- Calibration for most meter types
- OEM production calibrations
- NIST-traceable primary standards

NVLAP® Lab Code 200668-0

Note: NVLAP accreditation applies only to the Badger Meter Flow Dynamics calibration lab, located in Racine, Wis.

Control. Manage. Optimize.

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